

Convolution: slide, multiply, sum

step 5 of 9

input (5x5, shaded by value)

1	2	3	2	1
2	5	7	5	2
3	7	9	7	3
2	5	7	5	2
1	2	3	2	1

kernel: Gaussian blur / 16

1	2	1
2	4	2
1	2	1

output(1,1) =

weighted average of
the 9 window values

= 7

output (smoothed)

4.5	5.6	4.5
5.6	7	